OpenCV

OpenCV stands for **Open Source Computer Vision** and is a library of functions which is useful in real time computer vision application programming.The term Computer vision is used for a subject of performing the analysis of digital images and videos using a computer program

OpenCV-Python

OpenCV-Python is a Python wrapper around C++ implementation of OpenCV library. It makes use of NumPy library for numerical operations and is a rapid prototyping tool for computer vision problems. OpenCV-Python is a cross-platform library, available for use on all Operating System (OS) platforms including, Windows, Linux, MacOS and Android. OpenCV also supports the Graphics Processing Unit (GPU) acceleration.

Environment Setup

The command which is used to install pip is as follows −

```
pip install opencv-python
```

Performing this installation in a new virtual environment is recommended.

The current version of OpenCV-Python is 4.5.1.48 and it can be verified by following command −

```
>>> import cv2
>>> cv2.__version__
```

Reading an image

The **CV2** package (name of OpenCV-Python library) provides the **imread()** function to read an image.

The command to read an image is as follows −

```
img=cv2.imread(filename, flags)
```

The flags parameters are the enumeration of following constants –

.cv2.IMREAD_COLOR (1) – Loads a color image.
.cv2.IMREAD_GRAYSCALE (0) – Loads image in grayscale mode
.cv2.IMREAD_UNCHANGED (-1) – Loads image as such including alpha channel

The function will return an image object, which can be rendered using **imshow()** function. The command for using imshow() function is given below –

cv2.imshow(window-name, image)

The image is displayed in a named window. A new window is created with the AUTOSIZE flag set. The **WaitKey()** is a keyboard binding function. Its argument is the time in milliseconds.

The function waits for specified milliseconds and keeps the window on display till a key is pressed. Finally, we can destroy all the windows thus created.

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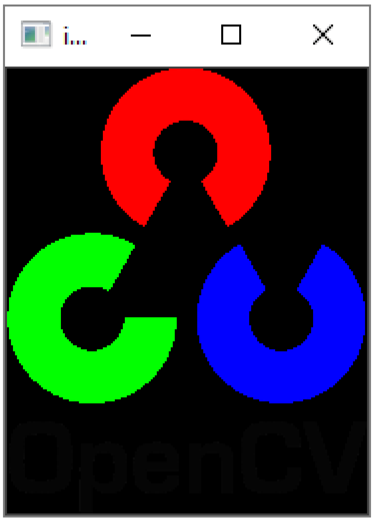
The program to display the OpenCV logo is as follows –

```
import numpy as np

import cv2

# Load a color image in grayscale

img = cv2.imread('OpenCV_Logo.png',1)

cv2.imshow('image',img)

cv2.waitKey(0)
```

```
cv2.destroyAllWindows()
```

The above program displays the OpenCV logo as follows −



Write an image

CV2 package has imwrite() function that saves an image object to a specified file.

The command to save an image with the help of imwrite() function is as follows −

cv2.imwrite(filename, img)

The image format is automatically decided by OpenCV from the file extension. OpenCV supports *.bmp, *.dib , *.jpeg, *.jpg, *.png,*.webp, *.sr,*.tiff, *.tif etc. image file types.

Example

Following program loads OpenCV logo image and saves its greyscale version when 's' key is pressed −

```python
import numpy as np

import cv2

# Load an color image in grayscale

img = cv2.imread('OpenCV_Logo.png',0)

cv2.imshow('image',img)

key=cv2.waitKey(0)

if key==ord('s'):

    cv2.imwrite("opencv_logo_GS.png", img)

cv2.destroyAllWindows()
```

Output

Using Matplotlib

Python's Matplotlib is a powerful plotting library with a huge collection of plotting functions for the variety of plot types. It also has imshow()

function to render an image. It gives additional facilities such as zooming, saving etc.

Example

Ensure that Matplotlib is installed in the current working environment before running the following program.

```
import numpy as np
import cv2
import matplotlib.pyplot as plt
# Load an color image in grayscale
img = cv2.imread('OpenCV_Logo.png',0)
plt.imshow(img)
plt.show()
```

Output

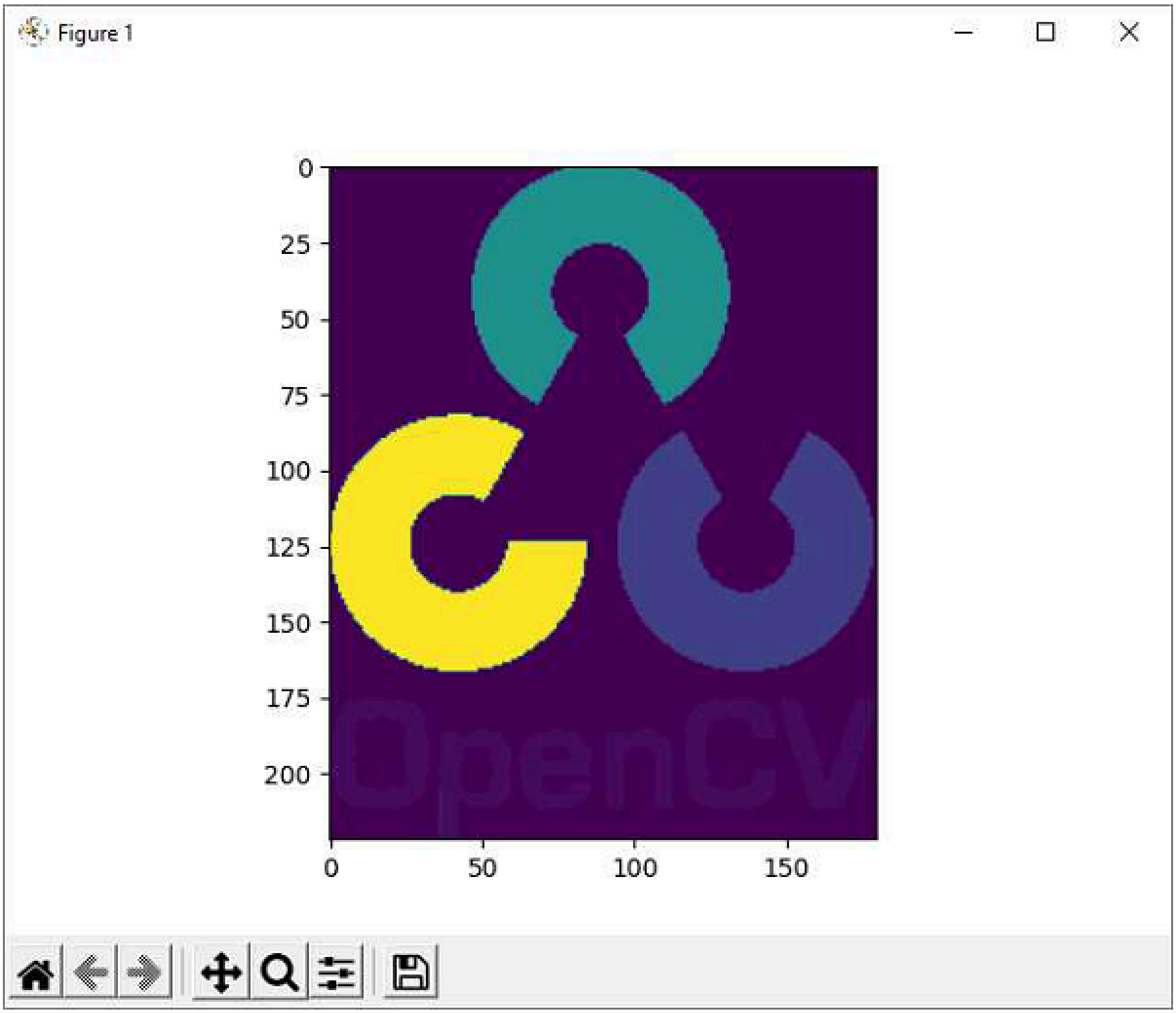


Image Properties

OpenCV reads the image data in a NumPy array. The **shape()** method of this ndarray object reveals image properties such as dimensions and channels.

The command to use the shape() method is as follows −

In the above command −

- The first two items shape[0] and shape[1] represent width and height of the image.
- Shape[2] stands for a number of channels.
- 3 indicates that the image has Red Green Blue (RGB) channels.

Similarly, the size property returns the size of the image. The command for the size of an image is as follows −

```
>>> img.size
```

```
119880
```